

30066 – MACROECONOMICS

(8 CFU)

CLASS : BIEM 14 & 19
2024 – 25

Lecture 23

(OB, Ch.18)

The open economy: the dictionary

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WHERE WE ARE

Fundamentals

1. Macroeconomic variables
 - Elements of national accounting
2. The short run
 1. Goods markets
 2. Financial markets
 1. Without commercial banks
 2. With commercial banks
 3. IS-LM model
3. The medium run
 1. Labor market
 2. Phillips curve
 3. IS-LM-PC model
 1. Medium and short run equilibrium
 2. Demand and supply shock

Extensions

4. Expectations, financial markets, and economic policies
 1. Financial markets and expectations
 2. Consumption, investment, and expectations
 3. Output, economic policies, and expectations
5. Open economy
 1. Exchange rates, trade, and international returns
 2. Goods markets in an open economy
 3. Financial markets and economic policies in an open economy
6. Public debt and fiscal policy
7. From the Great Recession to Quantitative Tightening
 - From a housing crisis to a financial crisis
 - Unconventional monetary policies



OPENNESS

When economies are open it is possible to make international exchange of:

1. Goods/services
2. Factors of production
3. Financial assets

Advantages and disadvantages:

- More variety and opportunities
- More competition and lower prices
- Interdependencies: shocks/policies in a country may affect other countries

Economies are becoming more interconnected...

Flows of goods, services, and finance, 1980–2014

\$ trillion, nominal

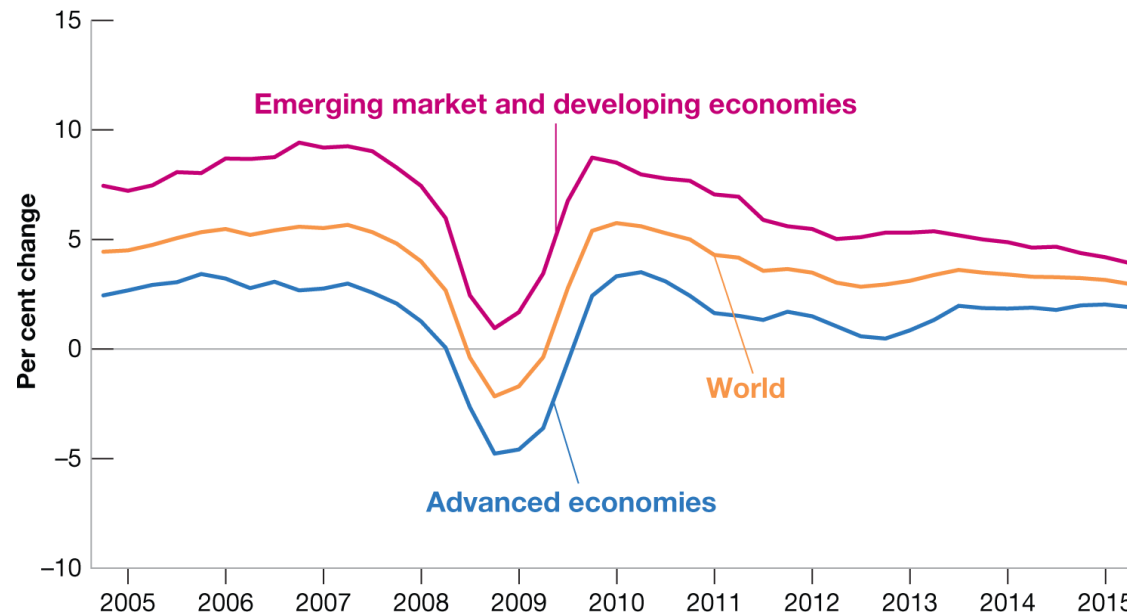
— All flows as % of GDP

- Finance
- Services
- Goods



SOURCE: UNCTAD; IMF Balance of Payments; World Bank; McKinsey Global Institute analysis

....which increases the risk of **contagion in the global economies**

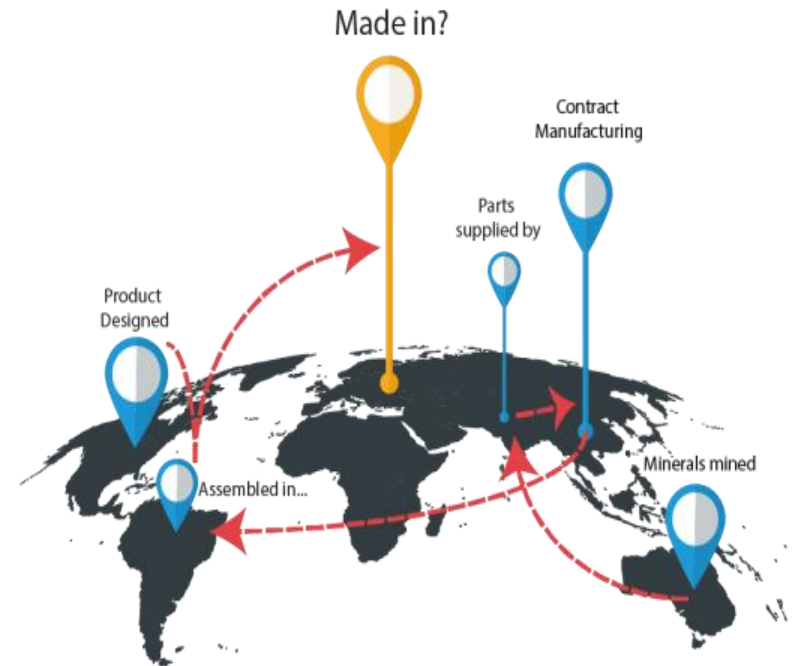


A crisis in one country can quickly spread to others through trade, financial markets, and investment flows

- The 2007-2009 crisis started in the US, but then diffused at the global level
- Covid-19 has highlighted the importance of international connections
- The war in Ukraine had an economic impact at the world level (energy prices, availability of goods, global value chains)

OPENNESS IN REAL ECONOMY

- Countries exchange goods/services (final and intermediate)
- The production process is broken across countries: firms specialize in a specific task and do not produce the whole product
 - Comparative advantage
 - Economies of scale and specialization
- Countries may affect the amount of trade imposing:
 - Tariffs/Quotas (Direct taxes or limits on imported goods)
 - Non-tariff barriers (technical standards and regulations, bureaucratic procedures and customs delays, product labeling requirements, sanitary and phytosanitary measures (especially in food and agriculture))
- International cooperation to reduce barriers:
 - WTO
 - Bilateral and regional agreement (EU; NAFTA; TPP; Mercosur, ...)



OPENNESS IN REAL ECONOMY

Figure 2.10. Trade Policies in Historical Perspective

The pace of tariff reduction and the coverage of free trade agreements has slowed.

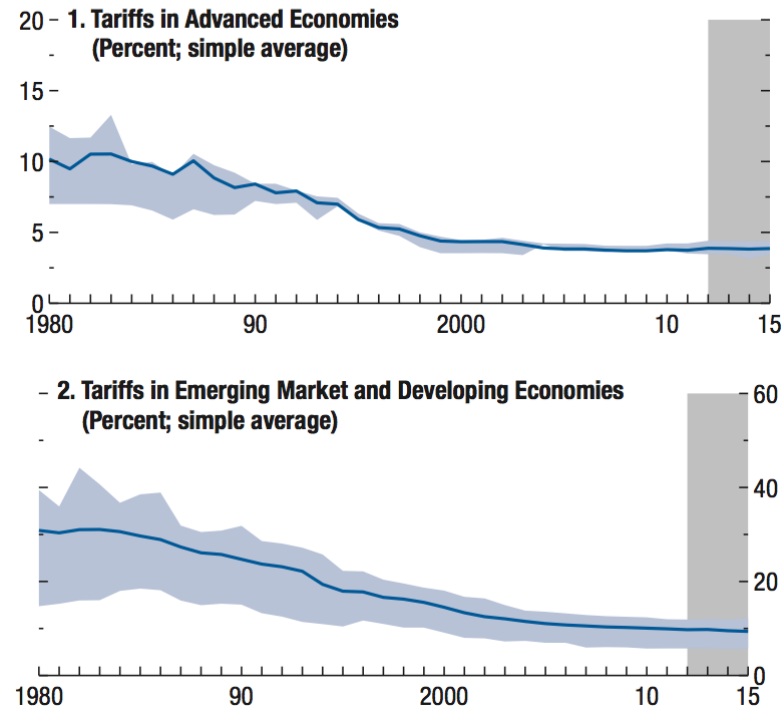
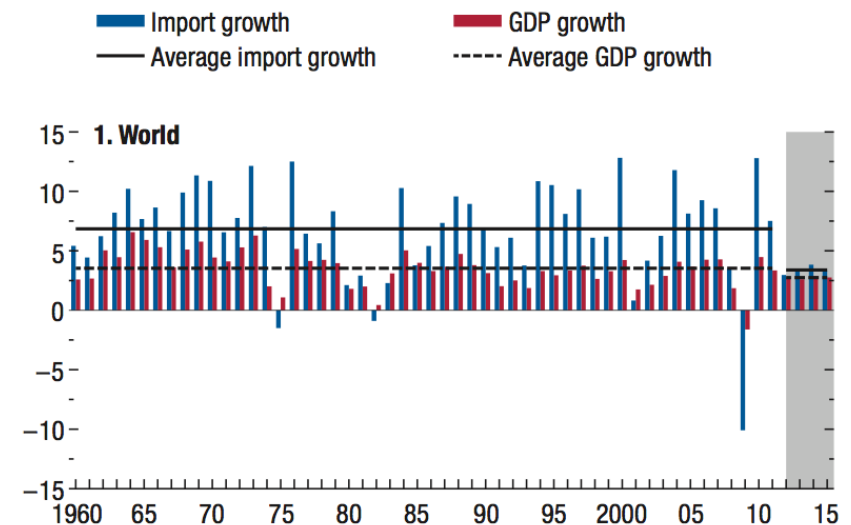


Figure 2.1. World Real Trade and GDP Growth in Historical Perspective

(Percent)

The decline in real trade growth since 2012 has been remarkable, especially when set against the historical relationship between growth in trade and global economic activity.



OPENNESS IN FINANCIAL MARKETS

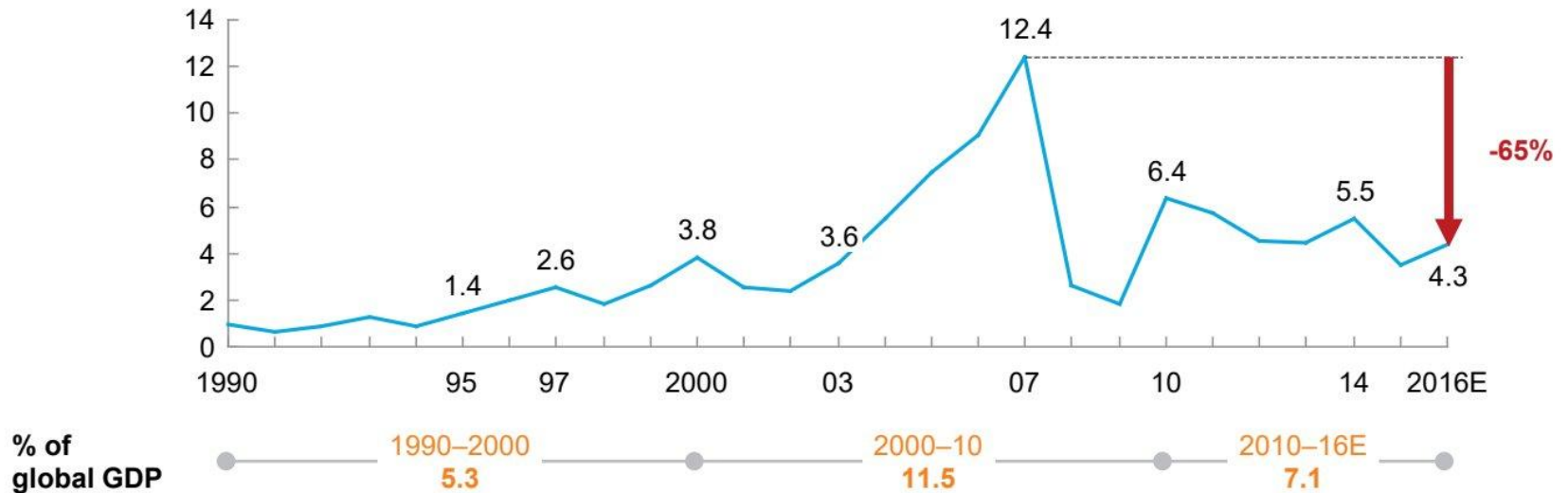
- Investors can choose between domestic and foreign assets
 - Based on relative expected returns
- Capital controls may be imposed with restrictions on:
 - Foreign assets residents can buy
 - Domestic assets foreigners can hold
- Financial markets are more open than goods markets
 - Easier and less costly to exchange financial assets than tangible goods



OPENNESS IN FINANCIAL MARKETS

Global cross-border capital flows¹

\$ trillion



¹ Gross capital inflows, including foreign direct investment (FDI), debt securities, equity, and lending and other investment.

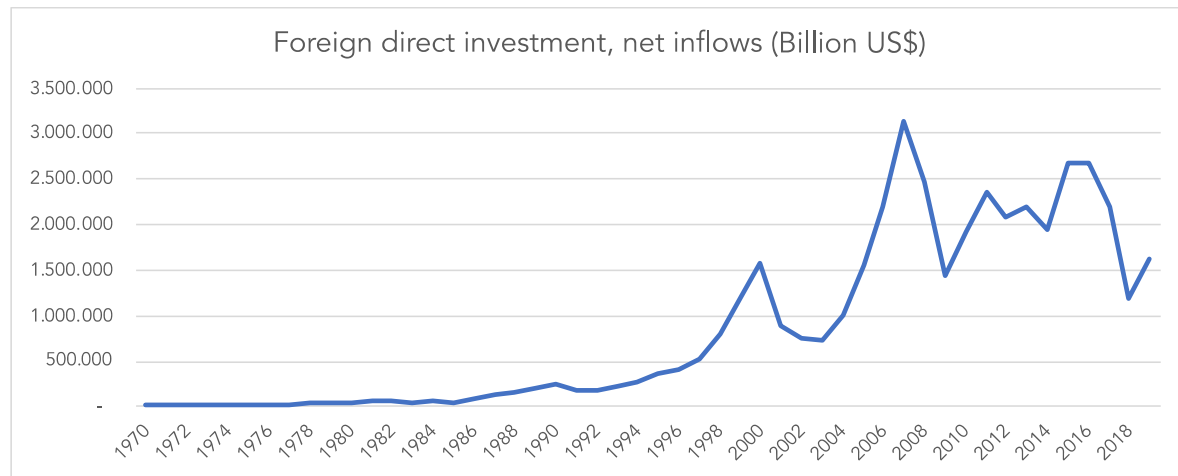
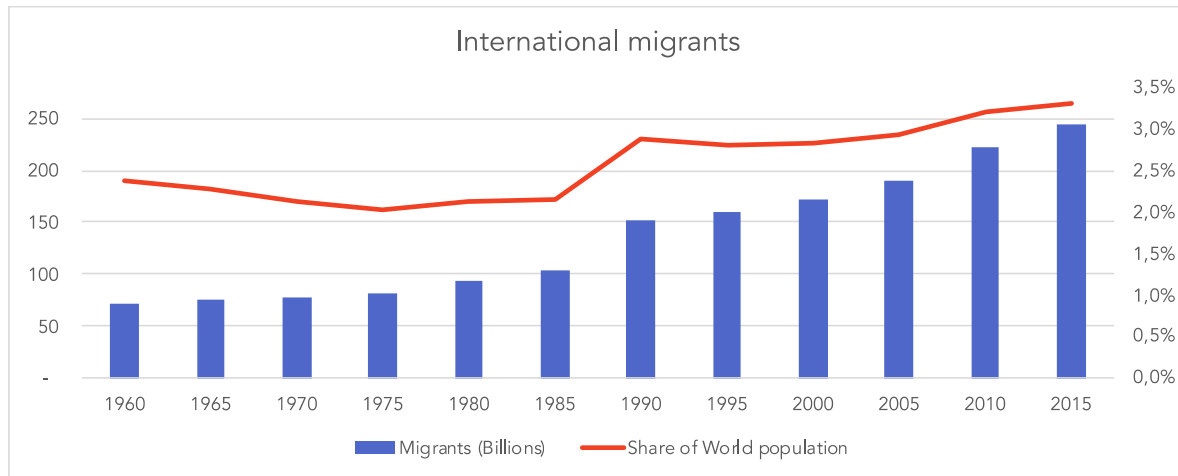
SOURCE: International Monetary Fund (IMF) Balance of Payments; McKinsey Global Institute analysis

Barriers: controls or restrictions on capital movements

FACTOR MARKET INTEGRATION

- To reduce costs and improve efficiency firms decide:
 - where to locate production (capital)
 - whether to use domestic or foreign labor
- Integration of factor markets means:
 - Capital and labor can move more freely across countries
- What are the effects
 - Wages: can increase or decrease depending on labor mobility
 - Employment: job opportunities may shift across countries
 - Investment flows : capital moves where returns are higher
 - Global supply chains: firms restructure production internationally

FACTOR MARKET INTEGRATION



Barriers: restrictions on the location of investments and migration policies

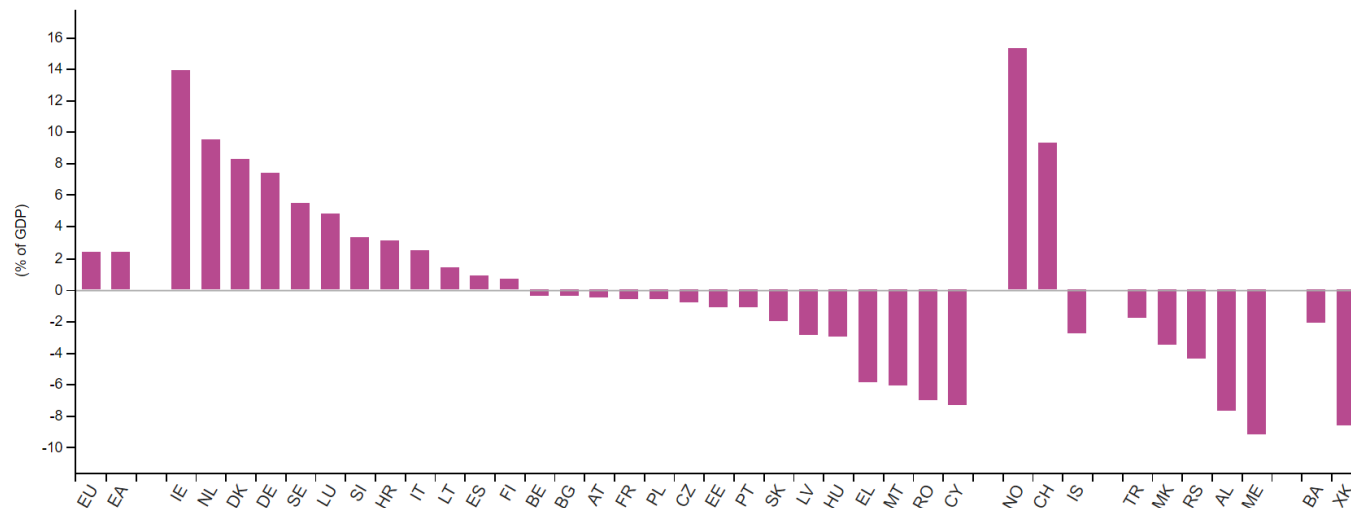
THE BALANCE OF PAYMENTS

[A nice video by Banque of France](#)

BALANCE OF PAYMENTS

- A set of accounts that summarizes a country's transactions with the rest of the world
- It is composed by three main accounts:
 1. **Current Account (CA)**: records payments to and from the rest of the world
 2. **Capital Account (KA)**: acquisition of non-financial assets by residents in the rest of the world, or by non-residents
 - $CA + KA$ = exposure of an economy to the rest of the world
 3. **Financial Account**: net changes in assets
 - FA = how it is financed

<https://www.imf.org/external/pubs/ft/bop/2007/bopman6.htm>



https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Balance_of_payment_statistics#Financial_account

CURRENT ACCOUNT

We will focus on the CA: the main determinants, how and why does it change

Current account (CA) records Exports (+) and imports (-) of goods/services

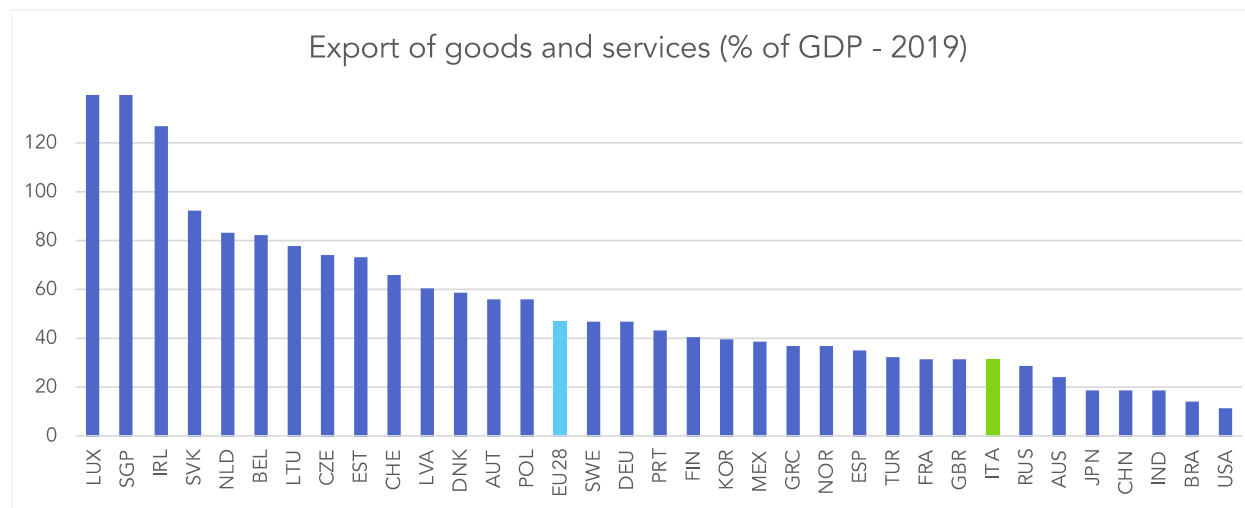
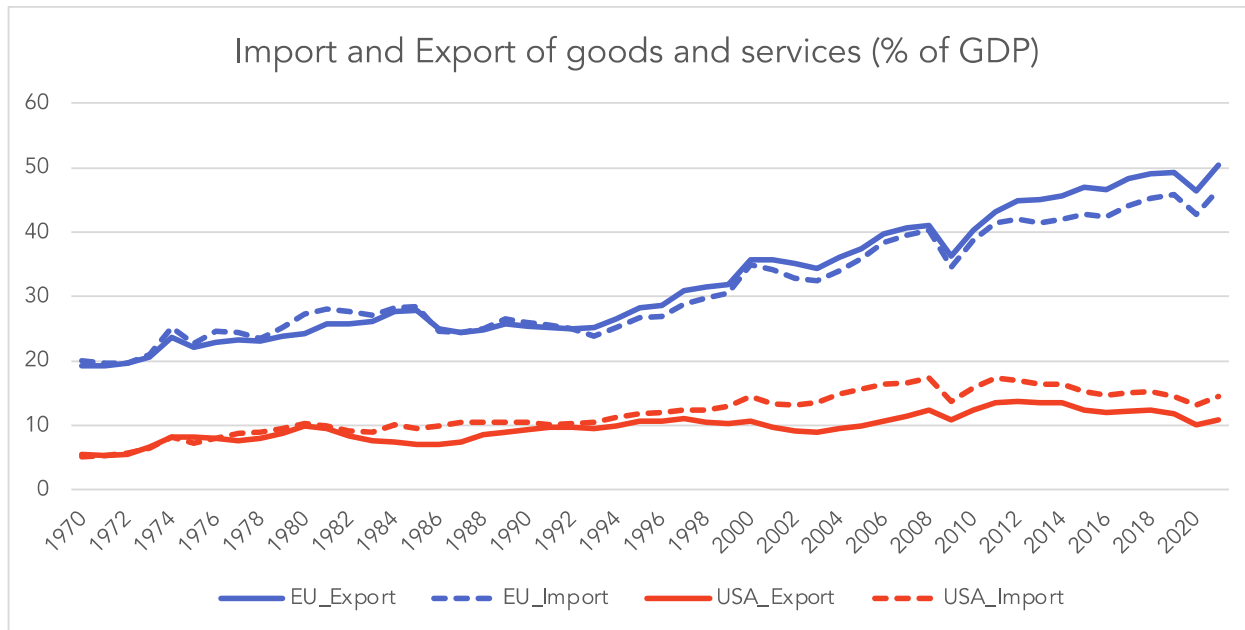
CA balance = net payments from abroad

CA>0: Surplus

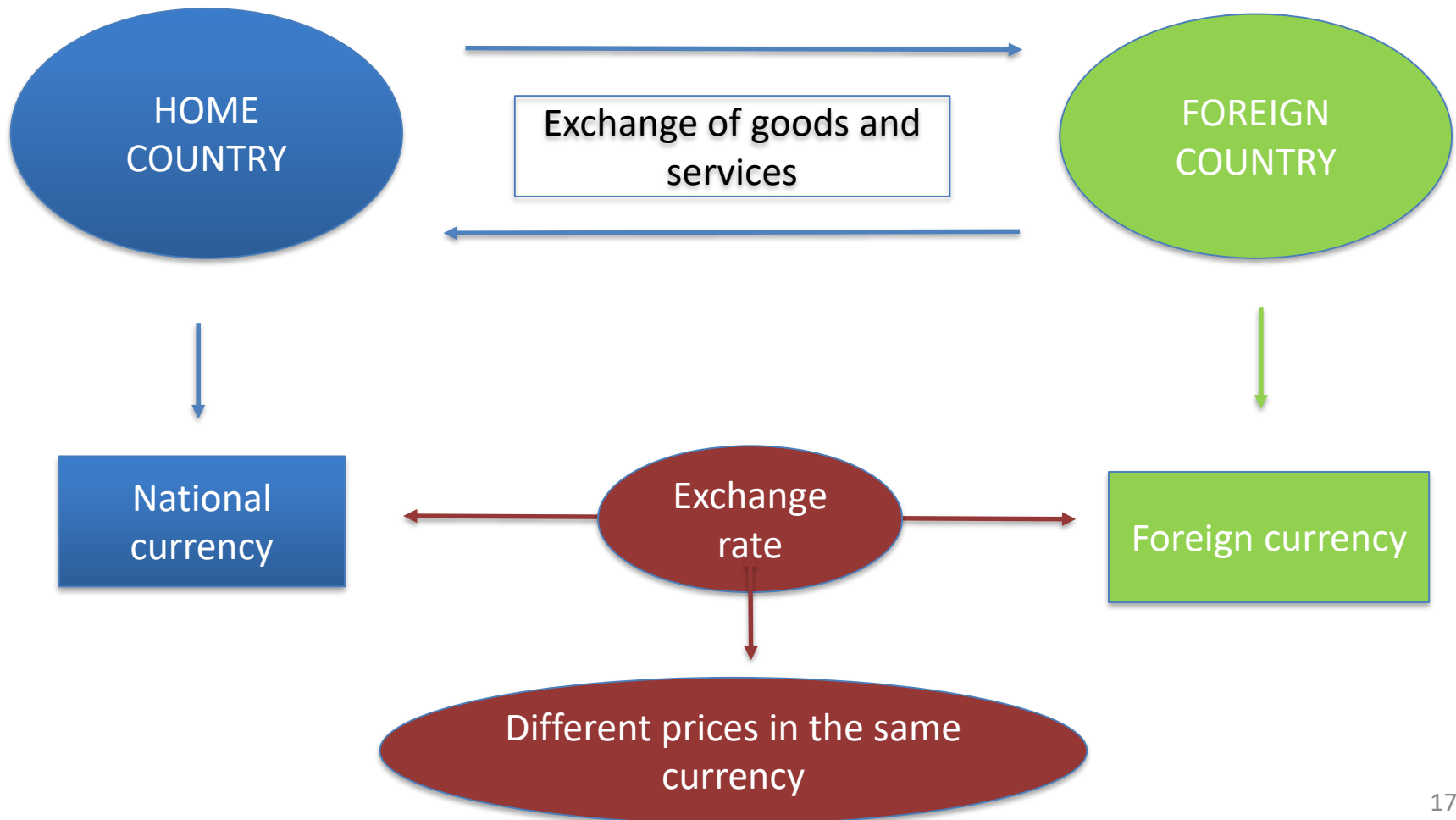
CA<0: Deficit

OPENNESS INDICATORS

- Level of imports/exports
- Imports and exports as a percentage of GDP
- Difference between exports and imports (trade balance)



Can X/GDP exceed 100%?



In an open economy, how do we compare domestic goods and foreign goods if each country has its own currency?

Using the relative price of the two currencies

E_t = the nominal exchange rate at time t = the ratio between the price of two currencies

$$E_t = \frac{\text{€}}{\text{\$}} \longrightarrow \text{How much is the euro worth today compared to the dollar?}$$

$$\text{If } E=1,1 \Leftrightarrow 1,1 = \frac{\text{€}}{\text{\$}} \Leftrightarrow 1,1\text{\$} = 1\text{€}$$

$\Rightarrow$ The price of 1 € is 1,1 \$

$\Rightarrow$ with 1 € we can buy 1,1 \$

$\Rightarrow$ it takes 1,1 \$ to buy 1 € $\Leftrightarrow$ It takes $1/1,1 = 0,9$ € to buy 1 \$

- If $E > 1$: the € is stronger than the \$ $\Leftrightarrow$ the domestic currency is worth more than the foreign one
- If $E < 1$: the € is weaker than the \$ $\Leftrightarrow$ the domestic currency is worth less than the foreign one

As rules, E can be expressed in two ways:

- Direct quotation (domestic currency per unit of foreign currency):
price of the domestic currency in terms of foreign currency
- Indirect quotation (foreign currency per unit of domestic currency):
price of the foreign currency in terms of domestic currency

International conventions:

- The Euro (€) is always expressed with direct quotation (domestic currency per unit of foreign currency)
- The US Dollar (\$) is always expressed with indirect quotation (foreign currency per unit of domestic currency)
- The British Pound (£) is always quoted with direct quotation, except against the Euro

Convention of the textbook and the macroeconomics course: always direct quotation (domestic currency per unit of foreign currency) => €/ \$

How is the exchange rate set in the markets?

It depends on the exchange rate regime chosen by the country

FLOATING

E can freely fluctuate over time based on demand and supply (purchases and sales) of the two currencies in the foreign exchange market

→ If the D of a currency $\uparrow$, its price $\uparrow$ with respect to the price of the other currency $\Rightarrow E \uparrow$

→ Instantaneous adjustments in the market as for € and \$

FIXED

E is kept constant over time at a predetermined level (exchange rate parity), set by the monetary authorities

→ Demand and supply imbalances are corrected through interventions by the Central Bank (buying or selling foreign currency)

→ No automatic adjustment → Adjustment happens through Central Bank actions, not by market forces

Argentina pegged the exchange rate at 1 Peso per 1 US Dollar

1) FLOATING EXCHANGE RATE: E is the relative price of the domestic currency

the value of a currency depends on how desirable it is in the market for carrying out transactions

- At any moment in the foreign exchange market, currencies are being bought and sold
- The relative demand for the domestic currency (compared to the foreign one) is therefore determined by these market transactions

NOMINAL APPRECIATION

E ↑

If the relative D for the domestic currency ↑, its price ↑ with respect to the price of the other currency

- ⇒ with 1 unit of domestic currency we can buy more units of foreign currency
- ⇒ domestic currency becomes stronger/ is worth more

$$E_{\text{€}/\$} = 1 \Rightarrow E'_{\text{€}/\$} = 1.2$$

With 1€, instead of buying only 1 \$, I can now buy 1.2\$

NOMINAL DEPRECIATION

E ↓

If the D for the foreign currency ↑ or the D for the domestic currency ↓

- ⇒ The relative price of the domestic currency ↓
- ⇒ Fewer \$ are needed to buy 1 €
- ⇒ Domestic currency loses value in the exchange rate market

With 1€, I can now buy fewer \$

Euro/Dollar Exchange Rate

US dollar (USD)



11 April 2025

EUR 1 = USD 1.1346 +0.0264(+2.4%)

Change from 4 January 1999 to 11 April 2025

Min (26 October 2000)	Max (15 July 2008)	Average
0.8252	1.5990	1.1830

Select: EUR vs. USD

Period: 04/01/1999

12/04/2025

Zoom:

1m

6m

1y

10y

all



https://www.ecb.europa.eu/stats/policy_and_exchange_rates/euro_reference_exchange_rates/html/eurofxref-graph-usd.en.html

2) FIXED EXCHANGE RATE

The chosen exchange rate ($E = \bar{E}$, the parity) is maintained —→ The CB intervenes by adjusting the M^s btw two or more countries regardless of the relative demand and supply of domestic and foreign currency in the foreign exchange market

If the relative D for the domestic currency $\uparrow$, it would tend to appreciate
=> to prevent this the CB sells domestic currency and buys foreign currency
=> $M^s \uparrow$
=> domestic currency loses value
=> foreign currency gains values
=> $E \downarrow$ until it returns to the initial parity

If the relative D for the domestic currency $\downarrow$, E would tend to $\downarrow$ and the CB buys domestic currency and sells foreign currency
=> $M^s \downarrow$
=> domestic currency gains value against the foreign one
=> $E \uparrow$ until it returns to the initial parity
...But the CB must have sufficient reserves of foreign currency

When the parity is changed by the authorities we have

- Revaluation: parity is increased $\bar{E} \uparrow$
- Devaluation: parity is decreased $\bar{E} \downarrow$

	Floating Exchange Rate	Fixed Exchange Rate
Exchange Rate (E)	Varies freely	Constant (parity)
Adjustment Mechanism	Market forces	Central Bank intervention
Central Bank Role	No intervention	Buys/Sells currency

CONSUMING DOMESTIC OR FOREIGN GOODS?

It depends on

- Preferences (home bias)
- Relative price

- production costs
- nominal exchange rate expresses the rate at which currencies are exchanged



Real exchange rate

Trade flows are determined by the relative competitiveness in the goods markets, in terms of:

- Exports (X): goods and services produced domestically and consumed abroad
- Imports (IM): goods and services produced abroad and consumed domestically

Net commercial position of a country determines net exports

$$NX = X - IM$$



Surplus:

$$X > IM \Leftrightarrow NX > 0$$

Trade balance

$$X = IM \Leftrightarrow NX = 0$$

Deficit:

$$X < IM \Leftrightarrow NX < 0$$

How much we export and import depends on how competitive we are (in terms of quality and price)

Consumers are interested in understanding the rate at which goods are exchanged

- ⇒ The relative price of domestic goods compared to foreign goods
- ⇒ N. of units of foreign goods that can be purchased with one unit of domestic goods



Real exchange rate

Ex: 2€ = price of 1 Kg of italian oranges
 3\$ = price of 1 Kg of US oranges
 E = €/\$ = 1,1

Real exchange rate = $(1,1 \times 2) / 3 = 2,2 / 3 = 0,7$ → If I don't buy 1 kg of Italian oranges, I save \$2.20, but I can only buy 0.7 kg of US oranges with that money → Italian goods are cheaper (more convenient), while US goods are more expensive

P = general domestic price level (GDP deflator)

P* = general foreign price level (foreign GDP deflator)

Price of national goods in foreign currency

Price of foreign goods in foreign currency

$$\varepsilon = \frac{EP}{P^*}$$

Real exchange rate : The price of domestic goods in terms of foreign goods

↑ E, ↑ P, ↓ P*

If $\varepsilon \uparrow$: Real Appreciation

Domestic goods become relatively more expensive than foreign goods, as more units of foreign goods are needed to purchase 1 unit of domestic goods

- ⇒ **lower international competitiveness** for the domestic country
- ⇒ ↓ **X**: Decrease in foreign demand for domestically produced goods
- ⇒ ↑ **IM**: Increase in domestic demand for foreign-produced goods

↓ E, ↓ P, ↑ P*

If $\varepsilon \downarrow$: Real Depreciation

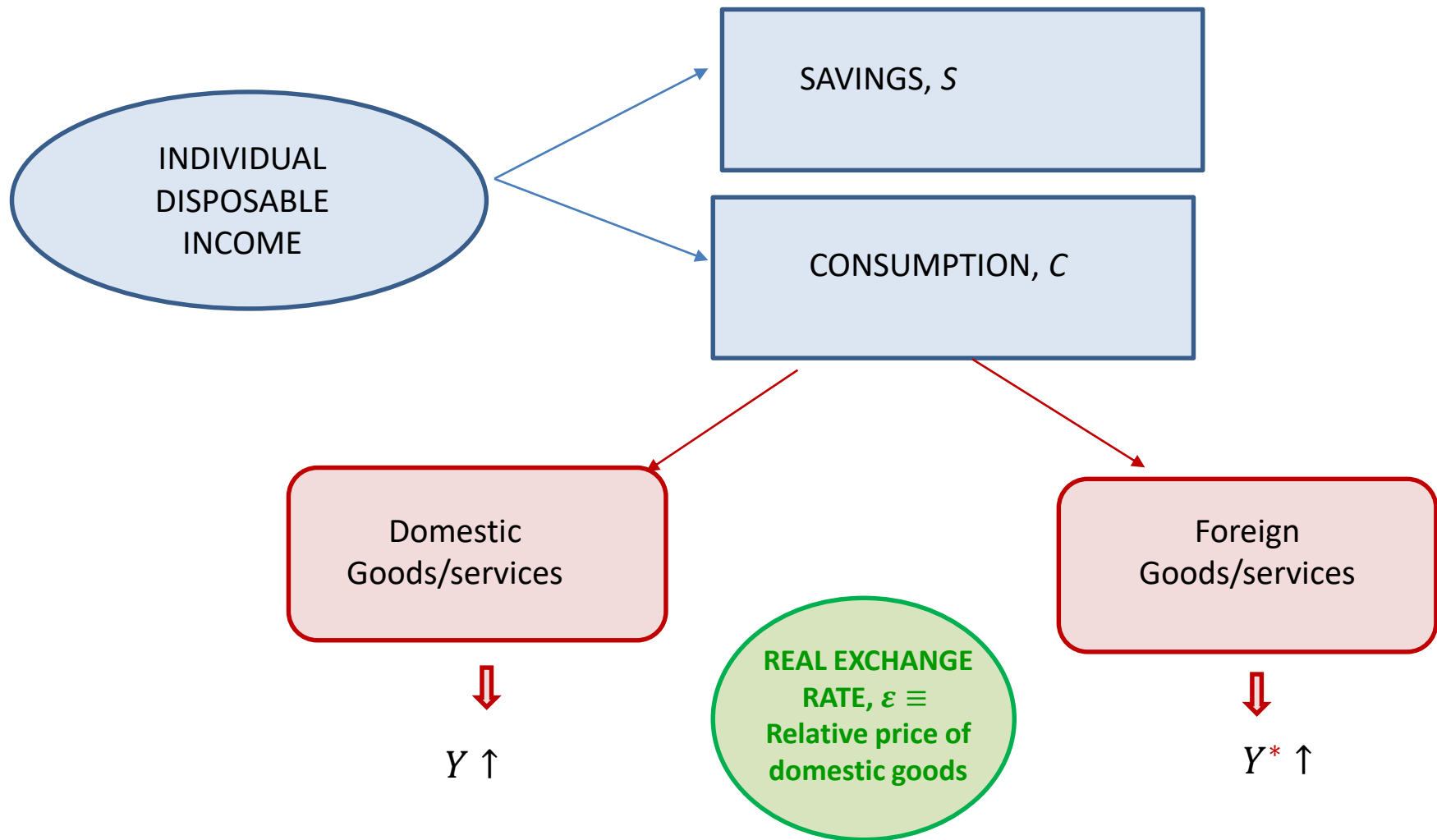
Domestic goods become relatively cheaper than foreign goods, as fewer units of foreign goods are needed to purchase 1 unit of domestic goods

- ⇒ **higher international competitiveness** for the domestic country
- ⇒ ↑ **X**: Increase in foreign demand for domestically produced goods
- ⇒ ↓ **IM**: Decrease in domestic demand for foreign-produced goods

OBSERVATIONS

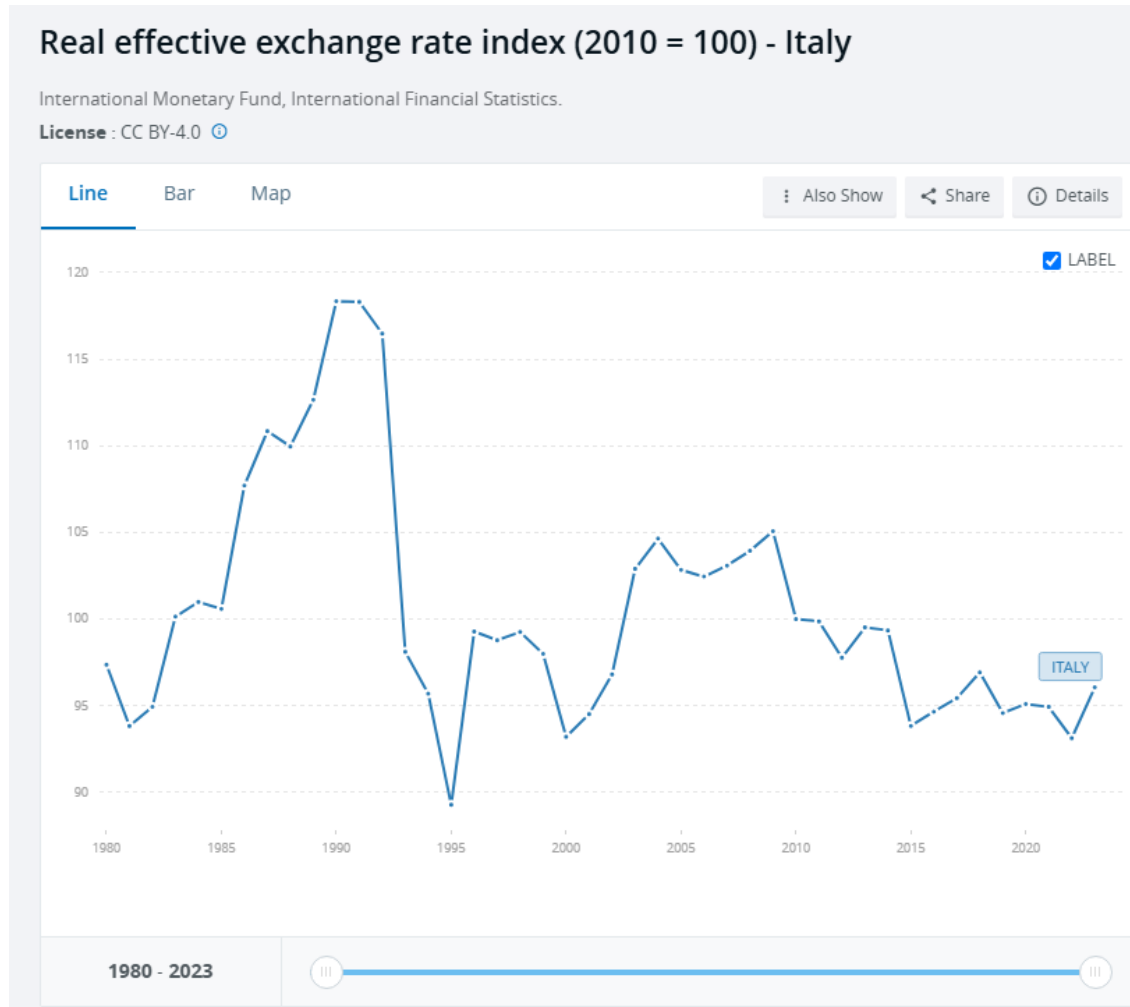
1. ϵ
 - is an indicator of competitiveness in international markets
 - is not directly observable but is instead constructed *ex post*
 - is an index number: its absolute level has no particular meaning => what matters are its changes over time, as these affect X and IM
2. **Bilateral exchange rate**: the real exchange rate between two specific countries (the one we have defined)
3. **Multilateral exchange rate**: measures the relative competitiveness of a country compared to multiple trading partners
 - P^* represents the weighted average of the prices of all the foreign countries the domestic country trades with
4. **REER (Real Effective Exchange Rate)**: the weighted average of a country's real exchange rate relative to an index or basket of major trading partners' currencies

Why do we care about X and IM?



Real Effective Exchange Rate Index

<https://data.worldbank.org/indicator/PX.REX.REER?locations=IT>



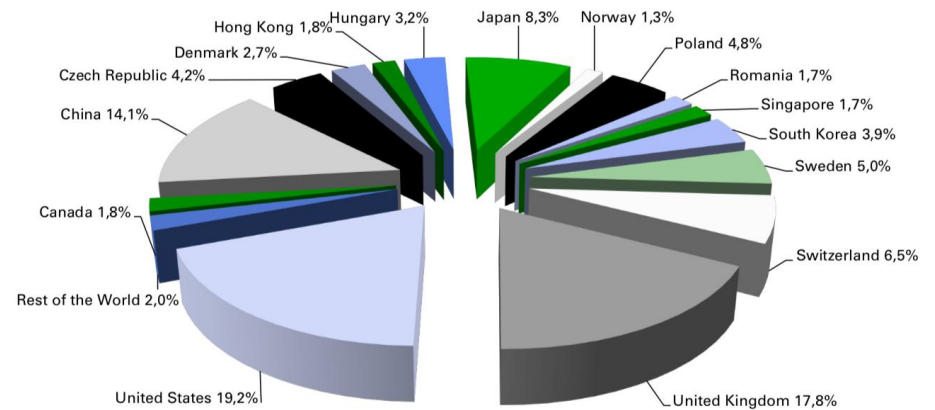
nominal effective exchange rate (a measure of the value of a currency against a weighted average of several foreign currencies) divided by a price deflator or index of costs

Euro Real Effective Exchange Rate Index

It is the weighted average of bilateral exchange rates

The weights are assigned based on:

- the share of trade with country i
- the degree to which country i competes in international trade with other countries



FINANCIAL MARKETS

The balance of payments must always be in equilibrium

$CA = -KA$ (+ statistical discrepancy)

What happens if a country:

- exports more than it imports?
 $CA > 0 \leftrightarrow KA$ deficit
- exports less than it imports?
 $CA < 0 \leftrightarrow KA$ surplus



What happens in real markets affects the financial market
 $\Rightarrow$ domestic agents must decide where to invest

How is an investment choice made in open economy?

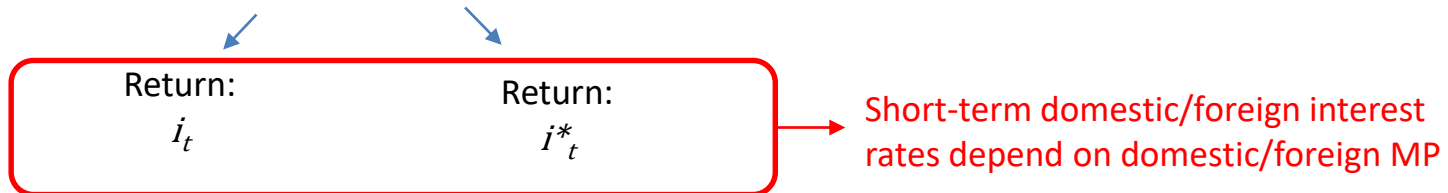
Agents can allocate their wealth btw domestic or foreign financial assets more/less risky



Based on their relative convenience:
Comparing expected return at a given maturity in the same currency

Assumptions:

- 1€ to invest for 1 year
- 2 options: 1-year domestic or foreign risk free bond



- E_t = current nominal exchange rate
- E_{t+1}^e = expected nominal exchange rate
- No transaction costs
- Agents risk neutral

t

$t+1$

Amount invested

Returns from holding the
asset for one year

1-year domestic bond

1€



$(1 + i_t) €$

Certain return in domestic
currency investing in
domestic assets

1-year foreign bond

1€

Buy E_t units
of foreign currency

$E_t \$$



$\frac{(1 + i_t^*)E_t}{E_{t+1}^e} €$

Expected return in
domestic currency investing
in foreign assets



With 1€ at time t , I can buy E_t units of foreign currency

⇒ by investing in foreign assets, in one year, I will have
 $(1 + i_t^*) E_t$ units of foreign currency

⇒ by converting this amount back into domestic
currency at the expected exchange rate, I expect to
have $\frac{(1+i_t^*)E_t}{E_{t+1}^e}$ units of domestic currency

By comparing these two values, the
individual agent decides where to invest

Where do we invest?

- If $(1 + i_t) \text{ €} > \frac{(1+i_t^*)E_t}{E_{t+1}^e} \text{ €}$

domestic bonds more attractive

- ⇒ on the exchange market investors demand more domestic currency and less foreign currency ($\text{€}^D \uparrow$)
- ⇒ the relative demand for domestic currency increases
- ⇒ Domestic currency appreciates
- ⇒ $E_t \uparrow$

- If $(1 + i_t) \text{ €} < \frac{(1+i_t^*)E_t}{E_{t+1}^e} \text{ €}$

foreign bonds more attractive

- ⇒ on the exchange market investors buy foreign currency and sell domestic currency ($\text{\$}^D \uparrow$)
- ⇒ the relative demand for domestic currency decreases
- ⇒ domestic currency depreciates
- ⇒ $E_t \uparrow$

In order to have equilibrium in financial markets and to trade both assets, the two returns must be equal on average

On average investors are indifferent between domestic and foreign bonds

$$(1 + i_t) \text{ €} = \frac{(1+i_t^*)E_t}{E_{t+1}^e} \text{ €}$$

UNCOVERED INTEREST RATE PARITY (UIP)

Arbitrage condition in the international financial markets

The UIP can be rewritten as

$$(i + i_t) = (1 + i_t^*) \frac{1}{\frac{E_{t+1}^e}{E_t} + 1 - 1}$$

$$(i + i_t) = (1 + i_t^*) \frac{1}{1 + \frac{E_{t+1}^e - E_t}{E_t}}$$

→ ΔE^e = expected appreciation (if >0) or depreciation (if <0) rate of the domestic currency btw t and t+1

=>
$$(i + i_t) = \frac{1 + i_t^*}{1 + \Delta E^e}$$
 → Exact relationship in international financial markets btw domestic and foreign nominal interest rates and the expected appreciation rate of the domestic currency

Through a logarithmic transformation, we obtain an approximate relationship



$$i_t \approx i_t^* - \Delta E^e$$

The domestic short-term nominal interest rate is approximately equal to the foreign one minus the expected appreciation rate of the domestic currency



$$i_t - i_t^* \approx -\Delta E^e$$

The difference btw the domestic and the foreign short-term nominal interest rate is approximately equal to the opposite of the expected appreciation rate of the domestic currency

OBSERVATIONS

- 1) In an open economy, financial investment choices depend on:

- Differential between domestic and foreign yield to maturity

$$i_t \geq < i_t^*$$

- Expectations of appreciation/depreciation

$$\Delta E^e$$

If $i_t > i_t^* - \Delta E^e \Leftrightarrow i_t - i_t^* > -\Delta E^e$
→ more convenient to invest on domestic assets

If $i_t = i_t^* - \Delta E^e \Leftrightarrow i_t - i_t^* = -\Delta E^e$
→ investors are indifferent btw domestic and foreign assets

If $i_t < i_t^* - \Delta E^e \Leftrightarrow i_t - i_t^* < -\Delta E^e$
→ more convenient to invest on foreign assets

- Risk (Greece 2015: huge increase in return + devaluation risk)
- Transaction costs: affect the relative convenience of where to invest by reducing the effective return

2) The UIP hold in all the exchange rate regime

FLOATING

E fluctuates over time based on purchases and sales of domestic and foreign currencies

$$\Delta E^e \neq 0$$



$$i_t \neq i_t^*$$

FIXED

$$E_t = \bar{E}$$

- If the chosen parity is believed by the markets

$$E_{t+1}^e = E_t = \bar{E} \Rightarrow \Delta E^e = 0$$



$$i_t = i_t^*$$

- If markets expect a currency devaluation or revaluation

$$E_{t+1}^e \neq E_t = \bar{E} \Rightarrow \Delta E^e \neq 0$$



$$i_t \neq i_t^*$$

QUICK CHECK

In a fixed exchange rate regime, what happens if there is excess demand for foreign currency?

- a) The Central Bank buys foreign currency to maintain the parity
- b) The domestic currency appreciates
- c) **The Central Bank sells foreign currency to maintain the parity**
- d) Interest rates automatically rise

QUICK CHECK

Which of the following is a necessary condition for maintaining a fixed exchange rate?

- a) Monetary policy independence
- b) Perfect capital mobility
- c) **Central Bank commitment to intervene**
- d) Floating interest rates

QUICK CHECK

In 1993, the nominal return on Brazilian government bonds was 36.5%, while the return on U.S. bonds was only 3%. However, the U.S. dollar was expected to appreciate by 34% against the Brazilian cruzeiro.

Where would it have been more convenient for an American investor to invest?

- a) In Brazilian bonds, because the return was significantly higher than in the U.S.
- b) In U.S. bonds, because the expected currency appreciation made the effective return on Brazilian bonds much lower**
- c) In Brazilian bonds, because exchange rate expectations are uncertain and shouldn't influence investment decisions
- d) It would make no difference—the returns would be equivalent on average

QUICK CHECK

Suppose that both a euro-denominated bond and a UK pound-denominated bond offer the same nominal return of 4%.

It would be more convenient to invest in the euro if

- a) **the euro is expected to appreciate against the pound**
- b) the euro is expected to depreciate against the pound
- c) the exchange rate is expected to remain stable
- d) the pound is expected to appreciate against the euro

QUICK CHECK

Suppose that a euro-denominated bond offers a nominal return of 2%, while a Japanese yen-denominated bond offers a nominal return of 3%.

It would be more convenient to invest in Japan if

- a) the euro is expected to appreciate against the yen by 1%
- b) the euro is expected to appreciate against the yen by 3%
- c) **the euro is expected to depreciate against the yen by 1%**
- d) the exchange rate is expected to remain stable

QUICK CHECK

Suppose that UIP holds. If the foreign interest rate is 2% and no depreciation of the domestic currency is expected over the next period, then the domestic interest rate will be

- a) lower than 2%
- b) higher than 3%
- c) impossible to say
- d) 2%**

QUICK CHECK

Suppose that UIP holds. If the interest rate in the US is 2% and the domestic currency is expected to appreciate by 3% over the next period, then the domestic interest rate will be:

- a) 5%
- b) 2%
- c) -1%**
- d) 1%

QUICK CHECK

If the interest rate in Japan is 1% and the domestic currency is expected to depreciate by 2% over the next period, where would it be more convenient for a domestic investor to invest?

- a) In Japanese bonds, because the interest rate is higher
- b) In domestic bonds, because the expected depreciation increases their effective return**
- c) In Japanese bonds, because the domestic currency depreciation reduces the return on domestic assets
- d) It makes no difference, returns are the same

TRUE OR FALSE

Suppose that the interest rate on Japanese bonds is 1% and the current €/¥ exchange rate is 100. Following a credibility crisis, financial markets begin to expect a 10% depreciation of the euro over the next year.

Assuming no transaction costs and no risk premium, and that the euro interest rate is 7% investor would prefer to invest in Japanese bonds.

FALSE

TAKE AWAY AND KEYWORDS

- Exchange rates
 - Nominal vs. real
- Exchange rate regimes
 - Fixed and floating
- Appreciation vs. depreciation
- Revaluation vs. devaluation
- Imports, exports, net export
- Interest rate differential
- Arbitrage condition
- UIP